

Numerical Methods for Differential Equations

Professor Marcus Grote

Dr. Romain Mottier

Fall semester 2026

Table of contents

I	Initial Value Problems (IVP) for Ordinary Differential Equations	3
I.1	Problem statement and examples	3
I.2	Existence and uniqueness theorems	5
I.3	Euler method (1768) and convergence of one-step methods	6
I.4	Explicit Runge-Kutta methods (1895–1901)	12
I.5	Adaptive Stepsize Control (SSC)	19
II	Multistep Methods	27
II.1	Adams Methods (~ 1850)	27
II.2	BDF Methods (Backward Differentiation Formulas)	32
II.3	Other Multistep Methods	34
II.4	Order of Multistep Methods	35

Initial Value Problems (IVP) for Ordinary Differential Equations

Contents

I.1 Problem statement and examples	3
I.2 Existence and uniqueness theorems	5
I.3 Euler method (1768) and convergence of one-step methods	6
I.4 Explicit Runge-Kutta methods (1895–1901)	12
I.5 Adaptive Stepsize Control (SSC)	19
I.5.1 Richardson Extrapolation (Lewis Fry Richardson, ~ 1920)	20
I.5.2 Embedded Runge-Kutta methods	25

I.1 Problem statement and examples

Example I.1 (Growth of a Population). Let $y(t)$ be the number of individuals in a population at time t .

$$\begin{cases} y'(t) = ay(t), & a \in \mathbb{R} \quad (\text{growth rate}) \\ y(0) = y_0. \end{cases}$$

The solution of this equation is $y(t) = e^{at}y_0$. One observes exponential growth of the population (see Figure I.1) for $a > 0$, or a decrease if $a < 0$.

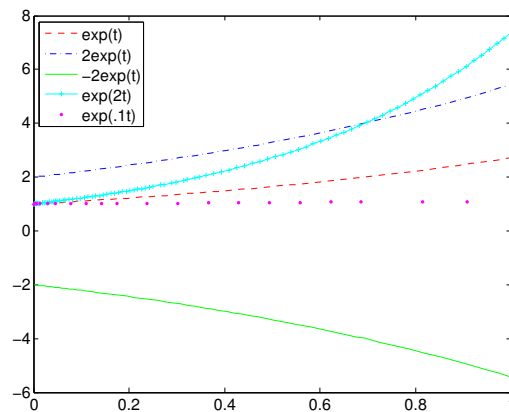


Fig. I.1: Exponential growth.

Example I.2 (Predator-Prey Model). Let

$$\begin{aligned} y_1(t) &= \text{number of prey (for example, rabbits) at time } t, \\ y_2(t) &= \text{number of predators (for example, lynxes) at time } t. \end{aligned}$$

$$\begin{cases} y_1' = (a - by_2)y_1, \\ y_2' = (cy_1 - d)y_2. \end{cases}$$

For given initial conditions $y_1(0)$ and $y_2(0)$, the solution $y: t \rightarrow \begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}$ describes the state of the system at every time t .

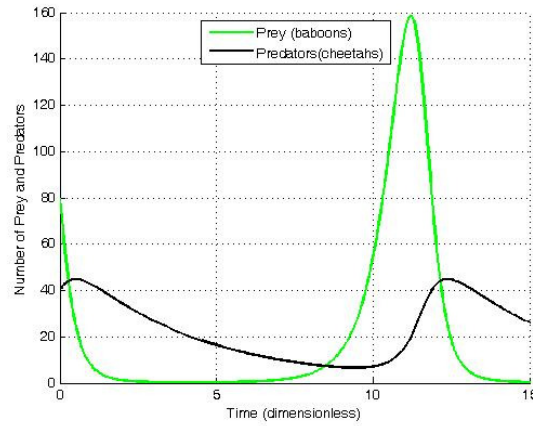


Fig. I.2: Population dynamics of cheetah and baboon.

Example I.3 (Displacement of a Mass). *The displacement from the equilibrium position at $y = 0$ is denoted by $y(t)$.*

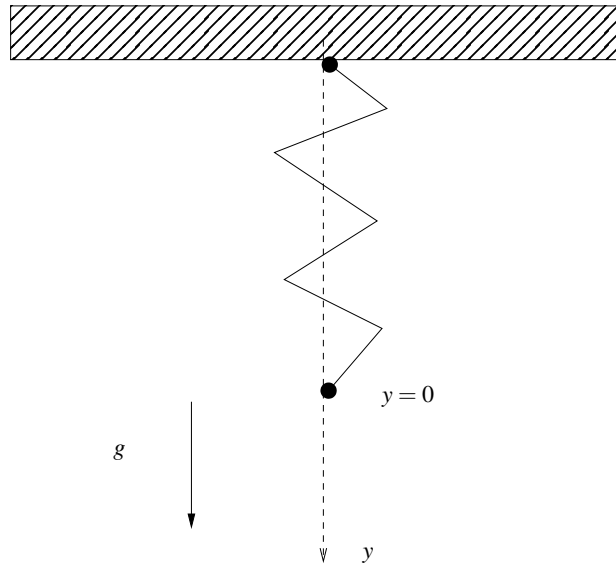


Fig. I.3: Displacement of a spring.

$$my''(t) = mg - ky(t). \quad (\text{I.1.1})$$

We set $y_1(t) := y(t)$ and $y_2(t) := y'(t)$ and rewrite (I.1.1) as a **first-order system of differential equations**:

$$\begin{cases} y_1' = y_2, \\ y_2' = g - \frac{k}{m}y_1. \end{cases}$$

In order for $y(t)$ to be uniquely determined, the initial position $y(0) = y_1(0)$ and the initial velocity $y'(0) = y_2(0)$ of the mass must be specified.

General Problem Statement. Given $y_0 \in \mathbb{R}^n$ and a function (“direction field”)

$$f: \begin{cases} U \rightarrow \mathbb{R}^n, \\ (t, y) \mapsto f(t, y), \end{cases} \quad \text{with } (t_0, y_0) \in U \subset \mathbb{R} \times \mathbb{R}^n \text{ open.}$$

We seek a function $y(t) \in \mathbb{R}^n$ on a time interval $t_0 \leq t \leq T$ which solves the **initial value problem (IVP)**,

$$\begin{cases} y' = f(t, y), \\ y(t_0) = y_0. \end{cases}$$

I.2 Existence and uniqueness theorems

Consider the IVP

$$\begin{cases} y' = f(x, y), \\ y(x_0) = y_0, \end{cases} \quad (x_0, y_0) \in U, \quad (\text{I.2.2})$$

with $f: U \rightarrow \mathbb{R}^n$ and $U \subset \mathbb{R} \times \mathbb{R}^n$ open.

Let $M = \max_D \|f(x, y)\|$ be the maximum speed in D . Let $\alpha = \min\{a, b/M\}$ be our work interval on which Peano theorem guarantees the existence of a solution.

Theorem I.1 (Existence: Peano, 1890). Let $D = \{(x, y) | x_0 \leq x \leq x_0 + a, \|y - y_0\| \leq b\} \subset U$ with $f|_D$ continuous,

$$M := \max_D \|f(x, y)\|, \quad \alpha = \min\{a, \frac{b}{M}\}.$$

Then there exists a solution $y(x) \in C^1([x_0, x_0 + \alpha], \mathbb{R}^n)$ of (I.2.2) with $y(x_0) = y_0$.

Proof. See “Ordinary Differential Equations”, Walter, Springer (p. 76) □

Theorem I.2 (Existence and Uniqueness: Cauchy-Lipschitz, 1824). Let $D = \{(x, y) | x_0 \leq x \leq x_0 + a, \|y - y_0\| \leq b\} \subset U$.

Assume

(i) $f|_D$ is continuous, $M = \max_D \|f(x, y)\|$, $\alpha := \min\{a, \frac{b}{M}\}$,

(ii) f is Lipschitz continuous with respect to y , meaning

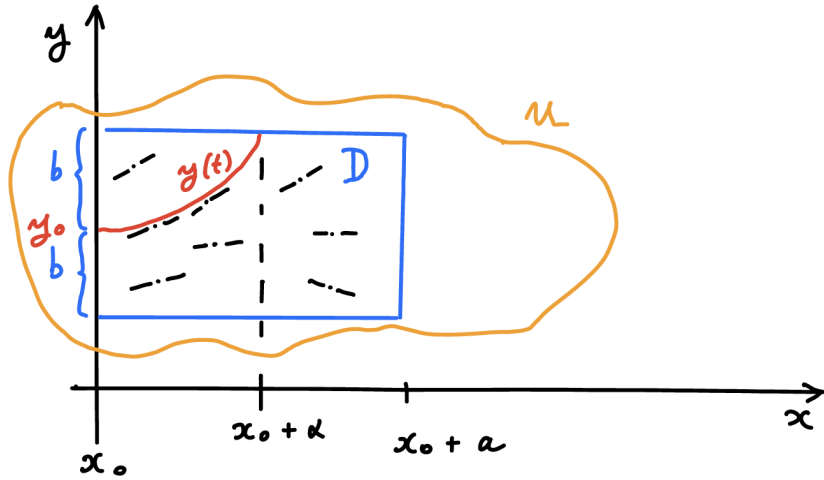


Fig. I.4: Local existence of a solution.

$\exists L > 0$ such that $\|f(x, y) - f(x, z)\| \leq L\|y - z\|$, $\forall (x, y), (x, z) \in D$.

Then there exists a unique solution $y(x)$ of (I.2.2) on $[x_0, x_0 + \alpha]$ with $y(x_0) = y_0$.

Proof. The modern proof uses the Banach fixed-point theorem (Picard iteration), which is why it is often called the Picard-Lindelöf theorem — see Analysis II. $\square$

Remark I.1.

1. $f(x, y) = \sqrt{|y|}$ is, for example, not Lipschitz continuous at $(0, 0)$ (see exercises).
2. If f is **not** Lipschitz continuous, the solution can still be unique (see exercises).
3. If $f \in C^1(D) \Rightarrow f|_D$ is Lipschitz continuous in y , because

$$\|f(x, y) - f(x, z)\| \underbrace{\leq}_{\text{Mean Value Theorem}} \underbrace{\max_D \|\partial_y f(x, y)\|}_{=L < \infty} \|y - z\|.$$

4. If $f \in C^k(D) \Rightarrow y \in C^{k+1}(D)$.

I.3 Euler method (1768) and convergence of one-step methods

Consider the IVP

$$\begin{cases} y' = f(x, y), \\ y(x_0) = y_0, \end{cases} \quad (\text{I.3.3})$$

with $f : U \rightarrow \mathbb{R}^n$ continuous and $(x_0, y_0) \in U \subset \mathbb{R} \times \mathbb{R}^n$ open.

1. **Goal:** A numerical approximation of $y(x)$ on the interval $[x_0, \bar{x}]$, for some fixed $\bar{x} > x_0$.

2. **Idea:** For $N \geq 1$, choose a step size $h := \frac{\bar{x} - x_0}{N} > 0$ and an equidistant partition of the interval $[x_0, \bar{x}]$ (with $x_0 < x_1 < \dots < x_N = \bar{x}$) with

$$x_i - x_{i-1} = h \quad \text{for } i = 1, 2, \dots, N.$$

For h sufficiently small, we approximate the derivative of the solution $y(x)$ at x_0 by the difference quotient

$$y'(x_0) \approx \frac{y(x_0 + h) - y(x_0)}{h} = \frac{y(x_0 + h) - y_0}{h}$$

and obtain

$$\underbrace{y(x_1) = y(x_0 + h)}_{\text{unknown}} \approx \underbrace{y_0}_{\text{known}} + h \underbrace{f(x_0, y_0)}_{\text{known}}.$$

Remark I.2. *These are exactly the first two terms of the Taylor expansion of the exact solution $y(x)$ at $x = x_0$.*

*Repeated application yields the **Euler Method**:*

$$y_{k+1} = y_k + hf(x_k, y_k), \quad k = 0, 1, \dots$$

Here, y_k is an approximation of the exact solution $y(x_k)$: $y_k \approx y(x_k)$ (see Figure I.5).

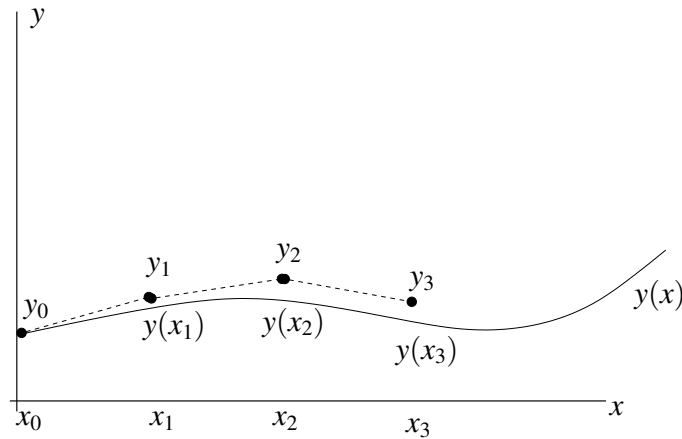


Fig. I.5: The Euler Method.

Remark I.3.

1. y_k depends on h . One also writes $y_k = y_h(x_k)$ to emphasize the dependence on h .
2. We already deviate from the exact solution after One step!

Example I.4. Consider the problem

$$\begin{cases} y'(t) = e^{y(t)} \sin(t) \\ y(0) = -\ln(3). \end{cases}$$

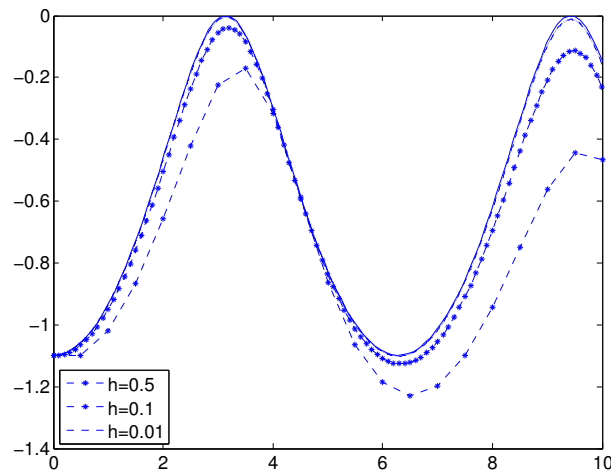


Fig. I.6: Euler method with step sizes $h = 0.5; 0.1; 0.01$.

Definition I.1. A method of the form

$$y_{k+1} = y_k + h \Phi(x_k, y_k, h), \quad k = 0, 1, \dots$$

with $x_{k+1} = x_k + h$ is called an **explicit one-step method** with increment function Φ .

Example I.5. The Euler method is an explicit one-step method with $\Phi(x_k, y_k, h) = f(x_k, y_k)$.

Definition I.2. Let $y(x)$ be the solution of the IVP (I.3.3) with $y(x_0) = y_0$. The local **discretization error** is defined as $y(x_0 + h) - y_1$, with $y_1 = y_0 + h \Phi(x_0, y_0, h)$. The method is called **consistent of (error) order p** if

$$y(x_0 + h) - y_1 = O(h^{p+1})$$

as $h \rightarrow 0$, for all IVPs with sufficiently differentiable $f(x, y)$.

The discretization error corresponds to the numerical error after only one step and is therefore also called the local error.

Example I.6. What is the error order p of the Euler method? A Taylor expansion gives

$$\begin{aligned} y(x_0 + h) - y_1 &= y(x_0) + \underbrace{y'(x_0)}_{=f(x_0, y_0)} h + y''(x_0) \frac{h^2}{2} + \mathcal{O}(h^3) - y_0 - hf(x_0, y_0) \\ &= \frac{y''(x_0)}{2} h^2 + \mathcal{O}(h^3) = \mathcal{O}(h^2), \quad h \rightarrow 0. \end{aligned}$$

Hence, the error order of the Euler method is $p = 1$.

Theorem I.3 (Convergence). Let $y(x)$ be the solution of the initial value problem

$$\begin{cases} y' = f(x, y), & x_0 \leq x \leq \bar{x} \\ y(x_0) = y_0 \end{cases}$$

on a fixed interval $[x_0, \bar{x}]$ with $f : U \rightarrow \mathbb{R}^n$ continuous, and let $\{y_k\}_{k=0}^N$ be the numerical solution of

$$\begin{cases} y_{k+1} = y_k + h \Phi(x_k, y_k, h), & k = 0, 1, \dots, N-1, \\ y_0 = y(x_0), & h = \frac{\bar{x} - x_0}{N}, \quad x_{i+1} - x_i = h \quad \forall i. \end{cases}$$

If

(a) the discretization error is consistent of order p , meaning

$$y(x+h) - y(x) - h \Phi(x, y(x), h) = \mathcal{O}(h^{p+1}), \quad h \rightarrow 0, \quad \forall x \in [x_0, \bar{x} - h]$$

(b) the function $\Phi(x, y, h)$ is Lipschitz continuous in y , meaning

$$\|\Phi(x, y, h) - \Phi(x, z, h)\| \leq \Lambda \|y - z\| \quad \forall (x, y), (x, z) \in U,$$

with $\{(x, y(x)) | x_0 \leq x \leq \bar{x}\} \subset U$, $U \subset \mathbb{R} \times \mathbb{R}^n$ open.

Then for the **global error** it holds

$$\|y(\bar{x}) - y_N\| \leq \underbrace{C \frac{e^{\Lambda(\bar{x}-x_0)} - 1}{\Lambda}}_{\text{constant}} h^p \quad h \rightarrow 0.$$

End of session 1

Remark I.4.

1. The global (“real”) error after N steps satisfies

$$\|y(\bar{x}) - y_N\| = \mathcal{O}(h^p) \rightarrow 0, \quad (h \rightarrow 0, N \rightarrow \infty).$$

Hence, a consistency error of order p implies convergence of order p .

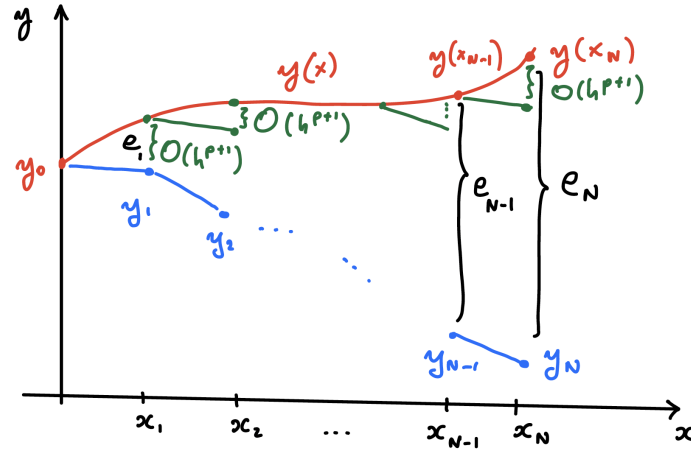


Fig. I.7: Global error in a one-step method.

2. The sequence $\{y_k\}_{k=0}^N$ also depends on h .

3. The size $\bar{x} - x_0 = hN$ remains constant during the limit process $h \rightarrow 0$, meaning $N \rightarrow \infty$.

Proof. First, assume that $[x_0, \bar{x}] \times \mathbb{R}^n \subset U$. Then the numerical solution is well-defined for all x_k , $h > 0$.

Let $e_k := y(x_k) - y_k$ be the (global) error after $k \leq N$ steps - see Figure I.7. We have $e_0 = y(x_0) - y_0 = 0$. Subtracting the following two equations

$$\begin{aligned} y(x_N) &= y(x_{N-1}) + h \frac{y(x_N) - y(x_{N-1})}{h} && \text{(trivial)} \\ y_N &= y_{N-1} + h \Phi(x_{N-1}, y_{N-1}, h) && \text{(Def. of 1-step method)} \end{aligned}$$

gives the error recursion $N-1 \mapsto N$:

$$\begin{aligned} e_N &= e_{N-1} + h \left[\frac{y(x_N) - y(x_{N-1})}{h} - \Phi(x_{N-1}, y_{N-1}, h) \right] \\ &= e_{N-1} + \underbrace{[y(x_N) - y(x_{N-1}) - h \Phi(x_{N-1}, y(x_{N-1}), h)]}_{\text{consistency term, } = O(h^{p+1})} \\ &\quad + \underbrace{h [\Phi(x_{N-1}, y(x_{N-1}), h) - \Phi(x_{N-1}, y_{N-1}, h)]}_{\text{controlled by Lipschitz continuity of } \Phi}, \end{aligned}$$

which holds not only for N but for all $k \leq N$. Applying the triangle inequality ($\|a + b\| \leq$

$\|a\| + \|b\|$) gives the following estimate for the global error as $h \rightarrow 0$:

$$\begin{aligned}
\|e_N\| &\leq (1 + h\Lambda)\|e_{N-1}\| + Ch^{p+1} \\
&\leq (1 + h\Lambda) \left[(1 + h\Lambda)\|e_{N-2}\| + Ch^{p+1} \right] + Ch^{p+1} \\
&\leq (1 + h\Lambda)^2\|e_{N-2}\| + Ch^{p+1}(1 + (1 + h\Lambda)) \\
&\vdots \\
&\leq (1 + h\Lambda)^N \underbrace{\|e_0\|}_{=0} + Ch^{p+1} \sum_{i=0}^{N-1} (1 + h\Lambda)^i
\end{aligned}$$

Computing the partial sum of the geometric series up to $i = N - 1$ and using $1 + x \leq e^x$, $\forall x$, we get

$$\begin{aligned}
\|e_N\| &\leq Ch^{p+1} \sum_{i=0}^{N-1} (1 + h\Lambda)^i \\
&\leq Ch^{p+1} \frac{(1 + h\Lambda)^N - 1}{1 + h\Lambda - 1} \\
&= Ch^p \frac{(1 + h\Lambda)^N - 1}{\Lambda} \\
&\leq Ch^p \frac{(e^{h\Lambda})^N - 1}{\Lambda} \\
&= Ch^p \frac{e^{hN\Lambda} - 1}{\Lambda} = \underbrace{C \frac{e^{\Lambda(\bar{x}-x_0)} - 1}{\Lambda}}_{\text{constant}} h^p.
\end{aligned}$$

Hence the estimate for the global error is proven.

Here, we assumed that Φ is defined and Lipschitz on the entire strip $[x_0, \bar{x}] \times \mathbb{R}^n$. In practice, Φ is often only defined in a neighborhood U of the solution, not everywhere. For large h , the numerical sequence can go outside of U .

Solution. In this case — here, for simplicity, $n = 1$ — consider for $x \in [x_0, \bar{x}]$ and $\varepsilon > 0$ small enough, so that the “ ε -strip” $\{(x, z) \mid x_0 \leq x \leq \bar{x}, |z - y(x)| \leq \varepsilon\} \subset U$, and define the following extension of Φ on $[x_0, \bar{x}] \times \mathbb{R}$:

$$\tilde{\Phi}(x, y, h) := \begin{cases} \Phi(x, y, h), & y \in [y(x) - \varepsilon, y(x) + \varepsilon] \\ \Phi(x, y(x) + \varepsilon, h), & y \geq y(x) + \varepsilon \\ \Phi(x, y(x) - \varepsilon, h), & y \leq y(x) - \varepsilon. \end{cases}$$

The extension $\tilde{\Phi}$ is now defined on all of $[x_0, \bar{x}] \times \mathbb{R}^n$ and is Lipschitz continuous. Moreover, the one-step method $\tilde{y}_{k+1} = \tilde{y}_k + h\tilde{\Phi}(x_k, \tilde{y}_k, h)$ is also consistent of order p .

$$\|y(x_k) - \tilde{y}_h(x_k)\| = \mathcal{O}(h^p), \quad \forall x_k \in [x_0, \bar{x}], \quad h \rightarrow 0.$$

Hence, there exists $h_0 > 0$ such that

$$\|y(x_k) - \tilde{y}_h(x_k)\| < \varepsilon, \quad \forall x_k \in [x_0, \bar{x}], \quad \forall h \in (0, h_0),$$

meaning $(x_k, \tilde{y}_k) \in U$, $\forall k \leq N$. But in U , $\Phi = \tilde{\Phi}$, so also $\tilde{y}_k = y_k$. Therefore, the above estimate holds for all $0 < h < h_0$. $\square$

Example I.7 (Euler Method). *If $f \in \mathcal{C}^1([x_0, \bar{x}] \times \mathbb{R}, \mathbb{R})$ with*

$\sup_{x_0 \leq x \leq \bar{x}, y \in \mathbb{R}} |\partial_y f(x, y)| \leq \Lambda$, then from the mean value theorem it follows that

$$|\Phi(x, y, h) - \Phi(x, z, h)| = |f(x, y) - f(x, z)| \leq \Lambda|y - z|, \quad \forall y, z \in \mathbb{R},$$

meaning if f is Lipschitz continuous in y , so is Φ . Since the Euler method is consistent of order $p = 1$, the theorem implies that it converges of order 1, meaning the error decreases linearly with h :

$$|y(x_N) - y_N| = \mathcal{O}(h) \rightarrow 0, \quad h \rightarrow 0.$$

Example I.8 (Euler method applied to the predator-prey model (Lotka-Volterra model)).

$$\begin{cases} y_1^{(k+1)} = y_1^{(k)} + h(a - by_2^{(k)})y_1^{(k)} \\ y_2^{(k+1)} = y_2^{(k)} + h(cy_1^{(k)} - d)y_2^{(k)}. \end{cases}$$

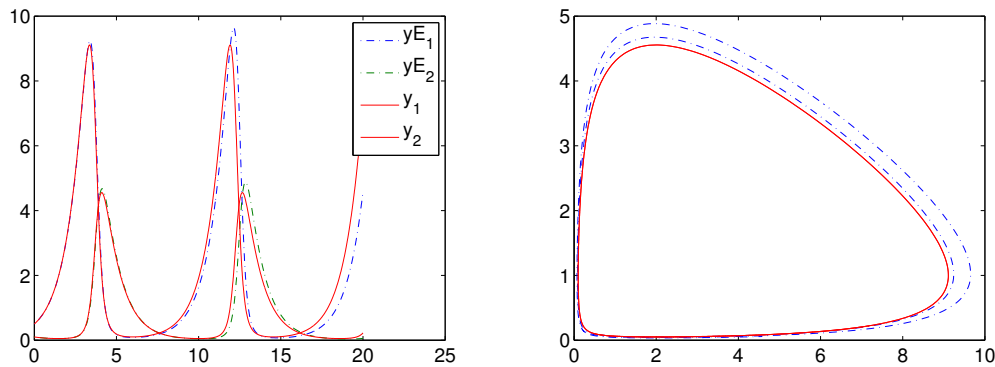


Fig. I.8: Numerical and exact solution (in red) for the Lotka-Volterra model: y_1, y_2 versus t (left); y_2 versus y_1 (right).

I.4 Explicit Runge-Kutta methods (1895–1901)

Goal. Find numerical methods of higher order

Idea. For the IVP

$$\begin{cases} y' = f(x) \\ y(x_0) = y_0, \end{cases} \quad (\text{I.4.4})$$

the Euler method applies the same rule once per subinterval:

$$\begin{aligned} y_1 &= y_0 + hf(x_0) \\ y_2 &= y_1 + hf(x_1) = y_0 + h(f(x_0) + f(x_1)) \\ &\vdots \\ y_N &= y_0 + h \sum_{i=0}^{N-1} f(x_i). \end{aligned}$$

On the reference interval $[0, 1]$, that rule reads $\int_0^1 f(x) dx \approx Q[f] = f(0)$ — a quadrature formula of order 1.

Runge Method (1895) (also called “modified Euler method”). The method corresponds to the midpoint quadrature formula $Q[f] = f(1/2)$ of order 2. But how one can even perform the first step $y_1 = y_0 + hf(x_0 + \frac{h}{2}, ?)$. To approximate $y(x_0 + h/2)$, we use an euler step of size $\frac{h}{2}$, meaning

$$y_1 = y_0 + h \underbrace{f\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}f(x_0, y_0)\right)}_{=: \Phi(x_0, y_0, h)}.$$

In practice it is more convenient to write the method in the following form:

$$\begin{cases} k_1 = f(x_0, y_0) \\ k_2 = f(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}k_1) \\ y_1 = y_0 + hk_2 \end{cases}$$

The Runge method has error order $p = 2$ (see exercises).

Heun Method (1900). The method corresponds to the trapezoidal quadrature formula $Q[f] = \frac{1}{2}(f(0) + f(1))$ of order 2:

$$y_N = y_0 + h \left[\frac{f(x_0)}{2} + \frac{f(x_0 + h)}{2} + \frac{f(x_1)}{2} + \frac{f(x_1 + h)}{2} + \dots + \frac{f(x_N)}{2} \right].$$

The method is given by

$$\begin{cases} k_1 = f(x_0, y_0) \\ k_2 = f(x_0 + h, y_0 + hk_1) \\ y_1 = y_0 + \frac{h}{2}[k_1 + k_2] \end{cases}$$

or

$$y_{k+1} = y_k + \underbrace{\frac{h}{2} (f(x_k, y_k) + f(x_k + h, y_k + hf(x_k, y_k)))}_{=h \Phi(x_k, y_k, h)}, \quad k = 0, 1, 2, \dots$$

Key question: *What is the **order of accuracy** of the Heun method?*

We expand the exact and numerical solutions after one step in h around x_0 :

1. **Exact solution.**

$$\begin{aligned} y(x_0 + h) &= y(x_0) + y'(x_0)h + y''(x_0)\frac{h^2}{2} + \mathcal{O}(h^3) \\ &= y(x_0) + hf(x_0, y_0) + \frac{h^2}{2} [\partial_x f(x_0, y_0) + \partial_y f(x_0, y_0)y'(x_0)] + \mathcal{O}(h^3) \end{aligned}$$

2. **Numerical solution.**

$$\begin{aligned} y_1 &= y_h(x_0 + h) = y_0 + \underbrace{\frac{h}{2} (f(x_0, y_0) + f(x_0 + h, y_0 + hf(x_0, y_0)))}_{=h \Phi(x_0, y_0, h)} \\ &= y_0 + \frac{h}{2} [2f(x_0, y_0) + \partial_x f(x_0, y_0)h + \partial_y f(x_0, y_0)f(x_0, y_0)h + \mathcal{O}(h^2)] \\ &= y_0 + hf(x_0, y_0) + \frac{h^2}{2} [\partial_x f(x_0, y_0) + \partial_y f(x_0, y_0)y'(x_0)] + \mathcal{O}(h^3). \end{aligned}$$

Subtracting the second from the first expression, we obtain

$$y(x_0 + h) - y_1 = \mathcal{O}(h^3), \quad h \rightarrow 0,$$

meaning the Heun method is consistent (at least) of order $p = 2$. By theorem I.3, it therefore converges with order 2, since Φ is Lipschitz continuous in y if f is.

This can now be generalized:

Definition I.3. Let $s \geq 1$ be an integer, $a_{ij}, b_i, c_i \in \mathbb{R}$ for $j = 1, \dots, i-1$, and $i = 1, \dots, s$, with $c_1 = 0$. The one-step numerical method

$$\begin{cases} k_1 &= f(x_0, y_0) \\ k_2 &= f(x_0 + c_2 h, y_0 + h a_{21} k_1) \\ &\dots \\ k_s &= f(x_0 + c_s h, y_0 + h \sum_{j=1}^{s-1} a_{sj} k_j) \\ y_1 &= y_0 + h \sum_{i=1}^s b_i k_i \end{cases}$$

is called a **s -stage explicit Runge-Kutta method**.

The (Butcher) tableau notation is often used:

0					
c_2	a_{21}				
c_3	a_{31}	a_{32}			
$\vdots$	$\vdots$	$\vdots$	$\ddots$		
c_s	a_{s1}	a_{s2}	$\dots$	$a_{s,s-1}$	
	b_1	b_2	$\dots$	b_{s-1}	b_s

Remark I.5. If an s -stage Runge-Kutta (RK) method is applied to the simple IVP $y' = f(x)$, $y(x_0) = y_0$, it corresponds to a quadrature formula with nodes c_i and weights b_i , $i = 1, \dots, s$:

$$y(x_0 + h) = y_0 + \int_{x_0}^{x_0+h} f(x) dx \simeq y_0 + h \sum_{i=1}^s b_i f(x_0 + c_i h)$$

The constants a_{ij} play no role in this case.

Example I.9.

Euler:

0		
1		1

Runge:

0		
1/2		1/2
		0 1

Heun:

0		
1		1
		1/2 1/2

4-stage Runge-Kutta method.

$$\left\{ \begin{array}{l} k_1 := f(x_0, y_0) \\ k_2 := f(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}k_1) \\ k_3 := f(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}k_2) \\ k_4 := f(x_0 + h, y_0 + hk_3) \\ y_1 := y_0 + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) \end{array} \right.$$

0				
1/2	1/2			
1/2	0	1/2		
1	0	0	1	
	1/6	2/6	2/6	1/6

This explicit one-step method has order $p = 4$ (without proof). Applied to $y' = f(x)$, it corresponds as a quadrature formula to Simpson's rule.

Example I.10. Consider the IVP

$$\begin{cases} y'(x) = e^{y(x)} \sin x, \\ y(0) = -\ln(3), \end{cases}$$

with exact solution $y(x) = -\ln(\cos(x) + 2)$.

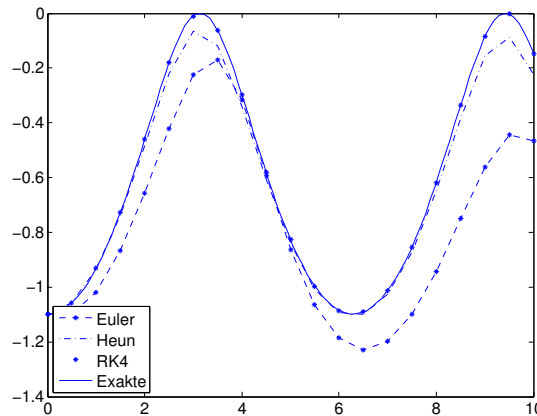


Fig. I.9: Exact and numerical solutions (Euler, Heun, RK4).

Lemma I.1. Consider an s -stage explicit Runge-Kutta method:

$$k_i = f(x_0 + c_i h, y_0 + h \sum_{j=1}^{i-1} a_{ij} k_j), \quad i = 1, \dots, s$$

$$y_1 = y_0 + h \sum_{i=1}^s b_i k_i$$

If $f(x, y)$ is Lipschitz continuous in y with Lipschitz constant L , then for

$\Phi(x, y, h) = \sum_{i=1}^s b_i k_i(x, y, h)$ it holds that

$$\|\Phi(x, y, h) - \Phi(x, z, h)\| \leq \Lambda \|y - z\|, \quad \forall y, z$$

with

$$\Lambda = L \left(\sum_{i=1}^s |b_i| + hL \sum_{i=1}^s \sum_{j=1}^{i-1} |b_i a_{ij}| + \mathcal{O}(h^2) \right),$$

meaning Φ is also Lipschitz continuous in y .

Proof. Step 1 (decompose Φ). Let

$$\Phi(x, y, h) = \sum_{i=1}^s b_i k_i(x, y, h), \quad k_i = f\left(x + c_i h, y + h \sum_{j=1}^{i-1} a_{ij} k_j\right)$$

and

$$\Phi(x, z, h) = \sum_{i=1}^s b_i \ell_i(x, z, h), \quad \ell_i = f\left(x + c_i h, z + h \sum_{j=1}^{i-1} a_{ij} \ell_j\right).$$

The triangle inequality gives

$$\|\Phi(x, y, h) - \Phi(x, z, h)\| = \left\| \sum_i b_i (k_i - \ell_i) \right\| \leq \sum_{i=1}^s |b_i| \|k_i - \ell_i\|,$$

Step 2 (stage 1). Directly from Lipschitz continuity of f :

$$\|k_1 - \ell_1\| = \|f(x, y) - f(x, z)\| \leq L \|y - z\|.$$

Step 3 (stage 2). Apply Lipschitz continuity of f again, this time to arguments that already differ by $h a_{21}(k_1 - \ell_1)$:

$$\begin{aligned} \|k_2 - \ell_2\| &= \|f(x + c_2 h, y + h a_{21} k_1) - f(x + c_2 h, z + h a_{21} \ell_1)\| \\ &\leq L \|y - z + h a_{21} (k_1 - \ell_1)\| \\ &\leq L (\|y - z\| + h |a_{21}| \|k_1 - \ell_1\|) \leq L(1 + h |a_{21}| L) \|y - z\|. \end{aligned}$$

Step 4 (general stage). At stage i the two arguments of f differ by $h \sum_{j<i} a_{ij}(k_j - \ell_j)$, so the same two ingredients (Lipschitz, then triangle inequality) give

$$\|k_i - \ell_i\| \leq L \left(\|y - z\| + h \sum_{j=1}^{i-1} |a_{ij}| \|k_j - \ell_j\| \right), \quad i = 1, \dots, s. \quad (\text{I.4.5})$$

We recognize a recursion: each stage is then bounded using the previous ones. Writing $A_i := \sum_{j<i} |a_{ij}|$ and substituting (I.4.5) into itself:

$$\begin{aligned} \|k_1 - \ell_1\| &\leq L \|y - z\|, \\ \|k_2 - \ell_2\| &\leq L(1 + h L A_2) \|y - z\|, \\ \|k_3 - \ell_3\| &\leq L(1 + h L A_3 + \mathcal{O}(h^2)) \|y - z\|, \quad \dots \end{aligned}$$

Each further substitution costs one more factor hL , so (see exercises for the full induction)

$$\|k_i - \ell_i\| \leq L(1 + \mathcal{O}(h))\|y - z\|, \quad i = 1, \dots, s,$$

where the $\mathcal{O}(h)$ starts with the explicit term hLA_i — which is all Step 5 needs.

Step 5 (collect powers of h). Ordering the Step 4 bound by powers of h ,

$$\|k_i - \ell_i\| \leq L\|y - z\| + hL^2A_i\|y - z\| + \mathcal{O}(h^2)\|y - z\|$$

with $A_i = \sum_{j=1}^{i-1} |a_{ij}|$, and injecting into the Step 1 bound gives, order by order in h ,

$$\|\Phi(x, y, h) - \Phi(x, z, h)\| \leq L\left(\sum_i |b_i| + hL \sum_i \sum_{j=1}^{i-1} |b_i a_{ij}| + \mathcal{O}(h^2)\right)\|y - z\| = \Lambda\|y - z\|.$$

□

Corollary I.1. *Let $y(x)$ be the solution of the IVP*

$$\begin{cases} y' = f(x, y) \\ y(x_0) = y_0 \end{cases}$$

with $f \in C^p(U, \mathbb{R}^n)$ and $U \supset \{(x, y(x)) \mid x_0 \leq x \leq \bar{x}\}$ open.

Then an (explicit) RK method of (local truncation) order p converges with order p , meaning

$$\|y(x_n) - y_n\| \leq Ch^p, \quad \forall x_n = x_0 + nh \in [x_0, \bar{x}], \quad h \rightarrow 0.$$

Proof. The RK method is a one-step method $y_{k+1} = y_k + h\Phi(x_k, y_k, h)$ with local truncation order p . From Lemma I.1 it follows that Φ is Lipschitz continuous in y . From Theorem I.3 it thus follows that convergence is $\mathcal{O}(h^p)$. □

Derivation of (explicit) high-order RK methods (ERK). One compares the Taylor expansions of $y(x_0 + h)$ and $y_1 = y_h(x_0 + h)$ and derives conditions on $b_i, c_i, a_{i,j}$ in order to achieve the highest possible order.

For example, if we require $p \geq 1$, then

$$\begin{aligned} y(x_0 + h) &= y_0 + hf(x_0, y_0) + \mathcal{O}(h^2) \\ y_h(x_0 + h) &= y_0 + h \sum_{i=1}^s b_i k_i = y_0 + h \sum_{i=1}^s b_i (f(x_0, y_0) + \mathcal{O}(h)) \\ &= y_0 + hf(x_0, y_0) \sum_{i=1}^s b_i + \mathcal{O}(h^2). \end{aligned}$$

Matching the first two terms of the two Taylor expansions above, meaning $p \geq 1$, requires $\sum_{i=1}^s b_i = 1$. This generalizes further.

Theorem I.4. *An s -stage RK method has order p . Then it holds that*

$$\sum_{i=1}^s b_i c_i^{q-1} = \frac{1}{q}, \quad q = 1, \dots, p. \quad (\text{I.4.6})$$

Proof. Case $p = 1$: see above.

Case $p \geq 2$: (see exercises) □

Remark I.6.

1. From (I.4.6), it follows that the corresponding quadrature formula with nodes $(0, c_2, c_3, \dots, c_s)$ and weights $(b_1, \dots, b_s)$ has at least order p : apply the quadrature formula to $f(x) = x^{q-1}$, $q \leq p$ (with $x_0 = y_0 = 0$, $h = 1$):

$$Q[f] = \sum_{i=1}^s b_i f(c_i) = \sum_{i=1}^s b_i c_i^{q-1} = \frac{1}{q} = \int_0^1 x^{q-1} dx, \quad q = 1, \dots, p.$$

Thus the quadrature formula is exact for all polynomials of degree $\leq p - 1$.

2. The condition (I.4.6) is necessary but not sufficient to achieve order p .
3. For $p \leq 4$, the minimal number of stages is $s = p$ (nothing forbids a method of order 4 with more stages). For $p \geq 5$, however, $s > p$ must hold! There is no explicit RK method with $p = s = 5$.
4. Arbitrarily high order p can be achieved, for example, by extrapolation methods based explicitly on Euler and corresponding to an ERK method with $s = 1 + p(p - 1)/2$. However, for $p = 10$, one then has $s = 46$, meaning 46 evaluations of f per step!
5. In practice, it is very complicated to derive explicit Runge-Kutta methods for large p (and as small s as possible). E. Hairer, for example, found in 1978 an ERK with $p = 10$ and $s = 17$.

I.5 Adaptive Stepsize Control (SSC)

Consider the initial value problem

$$\begin{cases} y' = f(x, y), & f : U \rightarrow \mathbb{R}^n, \\ y(x_0) = y_0, \end{cases}$$

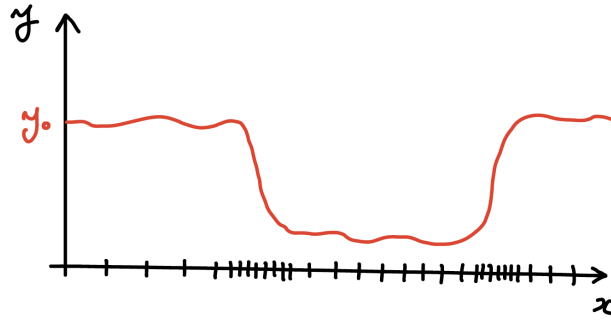


Fig. I.10: Adaptive stepsize control

Idea. Automatic adaptation of the stepsize h_i to the local behavior of the solution $y(x)$ – see Figure I.10.

Goal. For a given tolerance Tol, control the local error err per time step such that $\|err\| \leq \text{Tol}$. In doing so, we aim for:

1. **Efficiency:** Achieve a prescribed accuracy with as few steps as possible.
2. **Robustness:** Reliable selection of the (a priori) hard-to-predict “optimal” h_i .
3. **Problem:** Estimation of the local error – the exact solution $y(x)$ is unknown!
4. **Solution:** To estimate the local error, we compare two numerical solutions of different accuracy.

End of session 3

I.5.1 Richardson Extrapolation (Lewis Fry Richardson, ~ 1920)

Idea: Compare the numerical solution after two steps of length h with the numerical solution after one step of length $2h$.

$$\begin{array}{ll}
 y_1 &= y_0 + h\Phi(x_0, y_0, h) && \text{1st step of length } h \\
 y_2 &= y_1 + h\Phi(x_1, y_1, h) && \text{2nd step of length } h \\
 w &= y_0 + 2h\Phi(x_0, y_0, 2h) && \text{One step of length } 2h
 \end{array}$$

Let $z(x)$ denote the exact solution of the IVP with $z(x_1) = y_1$ – see Figure I.11.

We now assume that the numerical method has order $p \geq 1$, meaning

$$y(x_0 + h) - y_1 = C(x_0, y_0)h^{p+1} + \mathcal{O}(h^{p+2}).$$

where the leading coefficient $C(x_0, y_0)$ is independent of h (but depends on the starting point (x_0, y_0)).

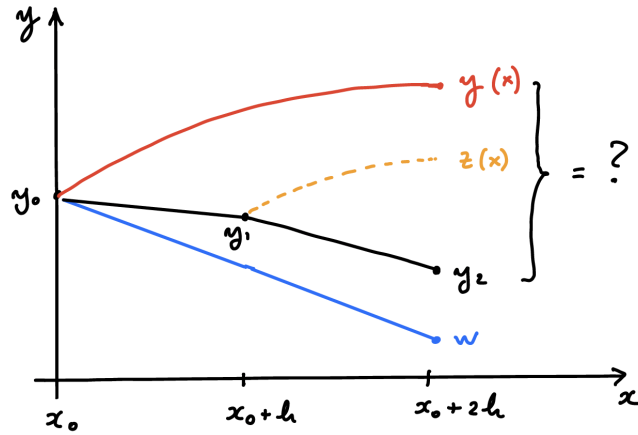


Fig. I.11: Richardson extrapolation

Remark I.7. For the Euler method, for example, we have (for sufficiently small h):

$$\begin{aligned} y(x_0 + h) - y_1 &= y''(x_0) \frac{h^2}{2} + \mathcal{O}(h^3) \\ &= \underbrace{\frac{1}{2} [\partial_x f(x_0, y_0) + \partial_y f(x_0, y_0) f(x_0, y_0)]}_{=C(x_0, y_0)} h^2 + \mathcal{O}(h^3) \end{aligned}$$

Goal. Estimate the local error after two steps, $y(x_0 + 2h) - y_2$.

Since $C(x_0, y_0)$ does not depend on h , we obtain, by applying the order- p relation at (x_0, y_0) with stepsize h , and again with stepsize $2h$:

$$y(x_0 + h) - y_1 = C(x_0, y_0) h^{p+1} + \mathcal{O}(h^{p+2}) \quad (*)$$

$$y(x_0 + 2h) - w = C(x_0, y_0) (2h)^{p+1} + \mathcal{O}(h^{p+2}) \quad (*')$$

We split the target error using $z(x)$:

$$y(x_0 + 2h) - y_2 = \underbrace{[y(x_0 + 2h) - z(x_0 + 2h)]}_{\text{error of the 1st step, transported}} + \underbrace{[z(x_0 + 2h) - y_2]}_{\text{error of the 2nd step}}.$$

We compute each bracket in turn.

Error of the second step. The step $y_1 \rightarrow y_2$ is just an ordinary one-step local error, starting from (x_1, y_1) with reference solution z . Applying $(*)$ at (x_1, y_1) gives

$$\begin{aligned} z(x_0 + 2h) - y_2 &= C(x_1, y_1) h^{p+1} + \mathcal{O}(h^{p+2}) \\ &= \underbrace{C(x_0 + h, y_0 + h\Phi(x_0, y_0, h))}_{C(x_0, y_0) + \mathcal{O}(h)} h^{p+1} + \mathcal{O}(h^{p+2}) \\ &= C(x_0, y_0) h^{p+1} + \mathcal{O}(h^{p+2}) \quad (**) \end{aligned}$$

Error of the first step. Now compare y and z at $x_0 + 2h$ by Taylor-expanding both around the base point $x_0 + h$:

$$\begin{aligned} y(x_0 + 2h) - z(x_0 + 2h) &\stackrel{\text{Taylor}}{=} \\ &y(x_0 + h) + hy'(x_0 + h) + \frac{h^2}{2}y''(x_0 + h) + \cdots - \underbrace{[z(x_0 + h) + hz'(x_0 + h) + \frac{h^2}{2}z''(x_0 + h) + \cdots]}_{=y_1} \\ = &[y(x_0 + h) - z(x_0 + h)] + h[y'(x_0 + h) - z'(x_0 + h)] + \frac{h^2}{2}[y''(x_0 + h) - z''(x_0 + h)] + \cdots \end{aligned}$$

We compare terms of equal powers of h in both Taylor expansions, the k -th one being

$$\frac{h^k}{k!} [y^{(k)}(x_0 + h) - z^{(k)}(x_0 + h)], \quad k = 0, 1, 2, \dots$$

- **Order 0:** we have $y(x_0 + h) - z(x_0 + h) = y(x_0 + h) - y_1$, which is exactly the first step's local error, $C(x_0, y_0)h^{p+1} + \mathcal{O}(h^{p+2})$ by (*). We call this the *order-0 gap*.
- **Order 1:** both y and z solve the same ODE, $y' = f(x, y)$ and $z' = f(x, z)$. Since $z(x_0 + h) = y_1$, this gives $z'(x_0 + h) = f(x_0 + h, y_1)$, while $y'(x_0 + h) = f(x_0 + h, y(x_0 + h))$. Subtracting,

$$\begin{aligned} y'(x_0 + h) - z'(x_0 + h) &= f(x_0 + h, y(x_0 + h)) - f(x_0 + h, y_1) \\ &= \partial_y f(x_0 + h, y(x_0 + h)) [y(x_0 + h) - y_1] + \underbrace{\mathcal{O}(h^{2(p+1)})}_{\subseteq \mathcal{O}(h^{p+2})}. \end{aligned}$$

- **Higher orders** ($k \geq 2$): same argument. Differentiating $y' = f(x, y)$ repeatedly, every derivative stays a function of (x, y) alone, built from f only:

$$y^{(k)} = g_k(x, y), \quad g_1 = f, \quad g_2 = \partial_x f + \partial_y f f, \quad \dots$$

With $z(x_0 + h) = y_1$, the order- k bracket is again one function at the *same* $x_0 + h$, at two y -values apart by the order-0 gap:

$$\begin{aligned} y^{(k)}(x_0 + h) - z^{(k)}(x_0 + h) &= g_k(x_0 + h, y(x_0 + h)) - g_k(x_0 + h, y_1) \\ &= \partial_y g_k(x_0 + h, y(x_0 + h)) \underbrace{[y(x_0 + h) - y_1]}_{\text{order-0 gap} = \mathcal{O}(h^{p+1})} + \underbrace{\mathcal{O}(h^{2(p+1)})}_{\subseteq \mathcal{O}(h^{p+1})} \\ &= \mathcal{O}(h^{p+1}), \end{aligned}$$

$$\implies \frac{h^k}{k!} [y^{(k)}(x_0 + h) - z^{(k)}(x_0 + h)] = \frac{h^k}{k!} \mathcal{O}(h^{p+1}) = \mathcal{O}(h^{k+p+1}) \subseteq \mathcal{O}(h^{p+3}) \quad (k \geq 2),$$

below the $\mathcal{O}(h^{p+2})$ we track: into the remainder.

Collecting the $k = 0, 1$ contributions and dropping $k \geq 2$ (shown negligible above) gives:

$$\begin{aligned}
 y(x_0 + 2h) - z(x_0 + 2h) &= [y(x_0 + h) - y_1] + h \partial_y f(x_0 + h, y(x_0 + h)) [y(x_0 + h) - y_1] + \mathcal{O}(h^{p+3}) \\
 &= \left\{ 1 + h \partial_y f(x_0 + h, y(x_0 + h)) \right\} [y(x_0 + h) - y_1] + \mathcal{O}(h^{p+3}) \\
 &= \underbrace{\left\{ 1 + h \partial_y f(x_0 + h, y(x_0 + h)) + \mathcal{O}(h^2) \right\}}_{\text{amplification factor} = 1 + \mathcal{O}(h)} \underbrace{[y(x_0 + h) - y_1]}_{C(x_0, y_0)h^{p+1} + \mathcal{O}(h^{p+2})} \\
 &= C(x_0, y_0)h^{p+1} + \mathcal{O}(h^{p+2}) \quad (***)
 \end{aligned}$$

Total error.

$$y(x_0 + 2h) - y_2 = \underbrace{y(x_0 + 2h) - z(x_0 + 2h)}_{(***)} + \underbrace{z(x_0 + 2h) - y_2}_{(**)} = 2C(x_0, y_0)h^{p+1} + \mathcal{O}(h^{p+2})$$

We now have two equations in the two unknowns $y(x_0 + 2h)$ and C :

$$y(x_0 + 2h) - w = C(2h)^{p+1} + \mathcal{O}(h^{p+2}) \quad (\text{I.5.7})$$

$$y(x_0 + 2h) - y_2 = 2Ch^{p+1} + \mathcal{O}(h^{p+2}) \quad (\text{I.5.8})$$

From (I.5.7)–(I.5.8) it follows that

$$y_2 - w = 2Ch^{p+1}(2^p - 1) + \mathcal{O}(h^{p+2})$$

which finally yields the desired estimate of the local error after two steps:

$$\Rightarrow \underbrace{y(x_0 + 2h) - y_2}_{\text{local error after 2 steps}} = \frac{y_2 - w}{2^p - 1} + \mathcal{O}(h^{p+2})$$

The leading term (for $h \rightarrow 0$) of the error estimator is fully computable, since y_2 and w are known. In addition, we obtain an improved numerical approximation $\hat{y}_2$,

$$y(x_0 + 2h) = \underbrace{y_2 + \frac{y_2 - w}{2^p - 1}}_{\hat{y}_2} + \mathcal{O}(h^{p+2})$$

of order $p + 1$.

Theorem I.5. Let $y(x)$ be the solution of the IVP $\begin{cases} y' = f(x, y), & x_0 < x < \bar{x} \\ y(x_0) = y_0 \end{cases}$ with $y \in C^{p+2}([x_0, \bar{x}])$, and let $y_1 = y_0 + h\Phi(x_0, y_0, h)$ be a one-step method of order $p \geq 1$.

Then, for $w = y_0 + 2h\Phi(x_0, y_0, 2h)$ and $y_2 = y_1 + h\Phi(x_1, y_1, h)$,

$$y(x_0 + 2h) - y_2 = \frac{y_2 - w}{2^p - 1} + \mathcal{O}(h^{p+2}), \quad h \rightarrow 0.$$

Moreover, $\hat{y}_2 := y_2 + \frac{y_2 - w}{2^p - 1}$ is an approximation of $y(x_0 + 2h)$ of order $p + 1$.

Proof. See above. □

Practical implementation.

1. **Given:** x_0, y_0, h, Tol

2. **Compute:** $y_2, w, err := \frac{\|y_2 - w\|}{2^p - 1} \approx \|y(x_0 + 2h) - y_2\|$

Choose here, for example, the max-norm $\|\cdot\|_\infty$.

If $err \leq Tol$, accept y_2 (or preferably even $\hat{y}_2$); otherwise repeat the step with the newly adapted step size $\hat{h}$.

Key question. *How can we adapt the step size to the local error?*

From

$$\begin{aligned} err(h) &= \bar{C}h^{p+1} = err \\ err(\hat{h}) &= \bar{C}\hat{h}^{p+1} = Tol \end{aligned}$$

it follows immediately that

$$\frac{Tol}{err} = \left(\frac{\hat{h}}{h}\right)^{p+1} \Rightarrow \hat{h} = h \left(\frac{Tol}{err}\right)^{1/(p+1)}$$

This is an estimate for the (new) “optimal” step size $\hat{h}$. Use $\hat{h}$ both for decreasing as well as for increasing the step size.

However, avoid overly abrupt changes: This yields the following algorithm, see Figure I.12.

Example I.11.

1. $y'(t) = -200ty^2$, $y(0) = 1$, $\Rightarrow y(t) = 1/(1 + 100t^2)$

2. *Brusselator (see exercises)*

$$\begin{aligned} y_1' &= 1 + y_1^2 y_2 - 4y_1 \\ y_2' &= 3y_1 - y_1^2 y_2 \end{aligned}$$

Remark I.8. *A disadvantage of Richardson extrapolation is that two steps are discarded if $err > Tol$.*

Algorithm ($\rho, \eta_{min}, \eta_{max}, Tol$ fixed)

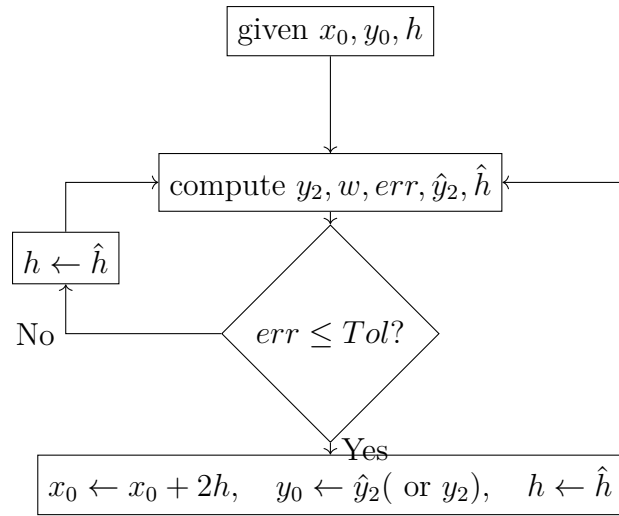


Fig. I.12: Adaptive stepsize control algorithm

I.5.2 Embedded Runge-Kutta methods

Objective. Estimate the error after only one step.

Idea. Use a second method $\hat{y}_1$ of order $\hat{p} < p$, often with $\hat{p} = p - 1$, see Figure I.13. From

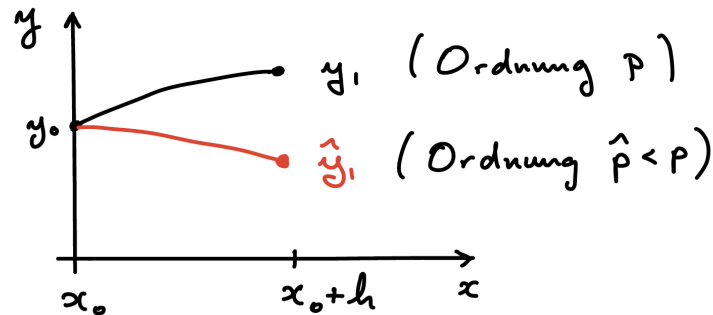


Fig. I.13: Embedded Runge-Kutta methods

$$\begin{aligned} y(x_0 + h) - y_1 &= Ch^{p+1} + \mathcal{O}(h^{p+2}) \\ y(x_0 + h) - \hat{y}_1 &= \hat{C}h^{\hat{p}+1} + \mathcal{O}(h^{\hat{p}+2}) \end{aligned}$$

it follows by subtraction that $y_1 - \hat{y}_1 = \hat{C}h^{\hat{p}+1} + \mathcal{O}(h^{\hat{p}+2}) \approx y(x_0 + h) - \hat{y}_1$.

As before, we can adapt h for the next step:

$$err := \|y_1 - \hat{y}_1\|_\infty \Rightarrow \hat{h} = \rho h \left(\frac{Tol}{err} \right)^{\frac{1}{\hat{p}+1}}$$

for example with $\rho = 0.9$

In order to keep the additional effort for $\hat{y}_1$ small, we use *the same stage derivatives* k_i as for y_1 :

$$\begin{aligned} y_1 &= y_0 + h \sum_{i=1}^s b_i k_i \\ \hat{y}_1 &= y_0 + h \sum_{i=1}^s \hat{b}_i k_i \end{aligned}$$

The error is then estimated by

$$err = \left\| h \sum_{i=1}^s (b_i - \hat{b}_i) k_i \right\|_{\infty}$$

Example I.12.

1. *Embedded Runge/Euler method* ($p = 2$):

$$\begin{aligned} k_1 &= f(x_0, y_0) \\ k_2 &= f\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2}k_1\right) \quad \Rightarrow \quad err := \|h(k_2 - k_1)\| \\ y_1 &= y_0 + hk_2 \quad \text{Runge method} \\ \hat{y}_1 &= y_0 + hk_1 \quad \text{Euler method} \end{aligned}$$

2. In Matlab, for example, `ode45` is based on the Dormand–Prince (1980) embedded RK method with $p = 5$ and $\hat{p} = 4$.

Multistep Methods

Consider the IVP $\begin{cases} y' = f(x, y), & f : U \rightarrow \mathbb{R}^N, \\ y(x_0) = y_0, \end{cases}$ with $U \subset \mathbb{R} \times \mathbb{R}^N$ open.

In a k -step method, the new value y_{n+1} at $x_{n+1} = x_n + h$ is determined from $f(x, y)$ and the previous $k \geq 1$ values, $y_n, y_{n-1}, \dots, y_{n-k+1}$.

Hope. *Achieve higher order with only one evaluation of f per step.*

Contents

II.1	Adams Methods (~ 1850)	27
II.2	BDF Methods (Backward Differentiation Formulas)	32
II.3	Other Multistep Methods	34
II.4	Order of Multistep Methods	35

II.1 Adams Methods (~ 1850)

Idea. The exact solution satisfies

$$y(x_{n+1}) = y(x_n) + \int_{x_n}^{x_{n+1}} f(t, y(t)) dt. \quad (\text{II.1.1})$$

- **Problem:** the integrand contains the unknown $y(t)$.
- **Solution:** replace f by the interpolation polynomial $p(t)$, $\deg p \leq k-1$, through the k known values

$$p(x_i) = f(x_i, y_i) =: f_i, \quad i = n, n-1, \dots, n-k+1, \quad x_{i+1} - x_i = h > 0 \quad \forall i.$$

Nodes span $[x_{n-k+1}, x_n]$, so on $[x_n, x_{n+1}]$ we integrate the **extrapolated** polynomial p exactly:

$$y_{n+1} = y_n + \int_{x_n}^{x_{n+1}} p(t) dt.$$

Remark II.1. *The values $y_n, \dots, y_{n-k+1}$ do not lie on the exact solution.*

For small k we can write down p by hand.

Case $k = 1$. One node, so $\deg p = 0 \Rightarrow p(t) = a$. As $p(x_n) = f_n$, it gives $a = f_n$, hence

$$p(t) = f_n = f(x_n, y_n) \quad \Rightarrow \quad y_{n+1} = y_n + \int_{x_n}^{x_{n+1}} f_n dt = y_n + hf_n \quad (\text{Euler method})$$

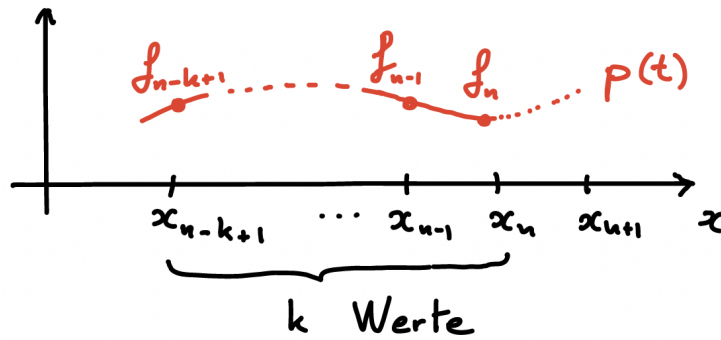


Fig. II.1: Interpolation of the k previous slopes in the AB- k method.

Case $k = 2$. Two nodes, so $\deg p \leq 1$. Therefore, $p(t) = a + b(t - x_n)$ and we impose the two interpolation conditions:

- $p(x_n) = f_n \Rightarrow a = f_n$;
- $p(x_{n-1}) = f_{n-1} \Rightarrow a + b(\underbrace{x_{n-1} - x_n}_{=-h}) = f_{n-1}$, hence $b = \frac{f_n - f_{n-1}}{h}$.

Therefore, $p(t) = f_n + (f_n - f_{n-1})\frac{t - x_n}{h}$ and by setting $t = x_n + sh$,

$$\Rightarrow p(t) = f_n + (f_n - f_{n-1})s, \quad t \in [x_n, x_{n+1}] \Leftrightarrow s \in [0, 1] \quad (\text{II.1.2})$$

Definition II.1. For $m \geq 0$, the m -th backward difference of f_n is defined as

$$\begin{aligned} \nabla^m f_n &= \nabla^{m-1} f_n - \nabla^{m-1} f_{n-1}, \quad m \geq 1 \\ \nabla^0 f_n &= f_n \end{aligned}$$

In general, one can show:

Proposition II.1 (Newton's Backward Difference Formula). *Let the nodes be equidistant with constant step $h > 0$, and measure t in steps from x_n :*

$$x_{n-i} = x_n - ih, \quad t = x_n + sh \quad \Leftrightarrow \quad s = \frac{t - x_n}{h},$$

so that $s = -i$ at the node x_{n-i} . Then the polynomial

$$p(t) = p(x_n + sh) = \sum_{j=0}^{k-1} \binom{s+j-1}{j} \nabla^j f_n$$

is the one interpolating f at the k nodes (x_{n-i}, f_{n-i}) , where $f_{n-i} := f(x_{n-i})$ for a function f of t , and $i = 0, \dots, k-1$.

Check ($k = 2$ nodes).

$$p(x_n + sh) = \underbrace{\binom{s-1}{0}}_{=1} \underbrace{\nabla^0 f_n}_{=f_n} + \underbrace{\binom{s}{1}}_{=s} \underbrace{\nabla^1 f_n}_{=f_n - f_{n-1}} = f_n + (f_n - f_{n-1})s,$$

which is exactly (II.1.2).

Proof. $k = 1, 2$ ✓

$k \geq 3$ (see exercises) □

Remark II.2.

$$1. \quad \binom{s+j-1}{j} := \underbrace{\frac{s(s+1)\dots(s+j-1)}{j!}}_{\text{polynomial of degree } j} \quad j \geq 1, \quad \binom{s-1}{0} := 1$$

$$2. \quad (-1)^j \binom{-s}{j} = \binom{s+j-1}{j}.$$

Now insert $p(t)$ from Proposition II.1 into (II.1.1):

$$\begin{aligned} y_{n+1} &= y_n + \int_{x_n}^{x_{n+1}} p(t) dt = y_n + h \int_0^1 p(x_n + sh) ds \\ &= y_n + h \int_0^1 \sum_{j=0}^{k-1} \binom{s+j-1}{j} \nabla^j f_n ds \\ &= y_n + h \sum_{j=0}^{k-1} \underbrace{\int_0^1 \binom{s+j-1}{j} ds}_{=: \gamma_j \text{ independent of } x_n, h, f_n, k} \nabla^j f_n \end{aligned}$$

For example, $\gamma_0 = \int_0^1 1 ds = 1, \quad \gamma_1 = \int_0^1 s ds = \frac{1}{2}$

j	γ_j
0	1
1	1/2
2	5/12
3	3/8
4	251/720
5	95/288
...	...

Adams–Bashforth (AB- k) methods.

$$y_{n+1} = y_n + h \sum_{j=0}^{k-1} \gamma_j \nabla^j f_n.$$

Using the table above for the coefficients γ_j , we can immediately determine the first AB- k methods:

- $k = 1$: $y_{n+1} = y_n + hf_n$ (explicit Euler)
- $k = 2$: $y_{n+1} = y_n + \frac{h}{2}(3f_n - f_{n-1})$ (AB-2)
- $k = 3$: $y_{n+1} = y_n + \frac{h}{12}(23f_n - 16f_{n-1} + 5f_{n-2})$ (AB-3)
- etc..

AB- k methods are explicit.

Remark II.3. *In each step only ONE new value f_n is computed, but the values $f_{n-1}, \dots, f_{n-k+1}$ must also be stored.*

Key question. *How does one compute the first $k - 1$ values $y_1, \dots, y_{k-1}$ for initialization when $k \geq 2$?*

Two options for the starting procedure

1. Classical one: use $k - 1$ RK steps, then continue with AB- k .
2. For variable step size methods: for small h , compute: $y_1 \leftarrow$ one-step method, $y_2 \leftarrow$ two-step method

Key question. *Now that we derived the Adams–Bashforth methods, what happens if we already knew $\mathbf{f}_{n+1}$?*

Let $p^*(t)$ be the polynomial of degree $\leq k$ interpolating $f_{n-k+1}, \dots, f_n, f_{n+1}$. Its newest node is now x_{n+1} , so Proposition II.1 must be applied with $k + 1$ nodes and the origin there:

$$\tilde{s} := \frac{t - x_{n+1}}{h} \quad \Longleftrightarrow \quad t = x_{n+1} + \tilde{s}h, \quad \text{so} \quad \tilde{s} = s - 1.$$

It follows that

$$p^*(t) = p^*(x_{n+1} + \tilde{s}h) = \sum_{j=0}^k \binom{\tilde{s} + j - 1}{j} \nabla^j f_{n+1}$$

$$\Rightarrow p^*(x_n + \underbrace{(\tilde{s} + 1)h}_s) = \sum_{j=0}^k \binom{s+j-2}{j} \nabla^j f_{n+1}$$

Insert $p^*(t)$ again into (II.1.1):

$$y_{n+1} = y_n + h \int_0^1 p^*(x_n + sh) ds = y_n + h \sum_{j=0}^k \underbrace{\int_0^1 \binom{s+j-2}{j} ds}_{=: \gamma_j^*} \nabla^j f_{n+1}$$

Adams–Moulton (AM- k) methods.

$$y_{n+1} = y_n + h \sum_{j=0}^k \gamma_j^* \nabla^j f_{n+1}$$

j	γ_j^*
0	1
1	$-1/2$
2	$-1/12$
3	$-1/24$
4	$-19/720$
...	...

Remark II.4.

1. To determine y_{n+1} we need to know $f_{n+1} = f(x_{n+1}, y_{n+1})$, which already contains y_{n+1} . Each step therefore requires the solution of a system in $\mathbb{R}^N$, nonlinear in general (linear only if f is linear in y) $\Rightarrow$ the price to pay to get higher convergence rates.
2. Therefore, AB- k methods are explicit, AM- k methods are implicit.

Using the table above for the coefficients γ_j^* , we can immediately determine the first AM- k methods:

- $k = 0$: $y_{n+1} = y_n + hf(x_{n+1}, y_{n+1})$ (implicit Euler)
- $k = 1$: $y_{n+1} = y_n + \frac{h}{2}(f_{n+1} + f_n)$ (trapezoidal method)
- $k = 2$: $y_{n+1} = y_n + \frac{h}{12}(5f_{n+1} + 8f_n - f_{n-1})$ (AM-2)
- etc..

Key question. How does one compute y_{n+1} ?

Collecting in the AM- k formula every term carrying f_{n+1} , and calling b everything else (it involves only y_n and $f_n, \dots, f_{n-k+1}$, all known):

$$y_{n+1} = h\beta f(x_{n+1}, y_{n+1}) + \underbrace{b}_{\text{known}}, \quad \beta := \sum_{j=0}^k \gamma_j^*.$$

It's nonlinear in y_{n+1} , so we solve it iteratively, here by a fixed-point iteration with iteration counter ℓ :

1. $y_{n+1}^{(0)} \leftarrow$ approximation, for example, by an AB- k method
2. ...
3. $y_{n+1}^{(\ell+1)} = h\beta f(x_{n+1}, y_{n+1}^{(\ell)}) + b, \quad \ell = 0, 1, 2, \dots$

If $f(x, y)$ is (locally) Lipschitz continuous in y with Lipschitz constant $L > 0$, then from

$$\begin{aligned} \|y_{n+1}^{(\ell+1)} - y_{n+1}^{(\ell)}\| &= h|\beta| \|f(x_{n+1}, y_{n+1}^{(\ell)}) - f(x_{n+1}, y_{n+1}^{(\ell-1)})\| \\ &\leq \underbrace{h|\beta|L}_{<1} \|y_{n+1}^{(\ell)} - y_{n+1}^{(\ell-1)}\| \\ &\leq \underbrace{(h|\beta|L)^\ell}_{\rightarrow 0 \quad (\ell \rightarrow \infty)} \|y_{n+1}^{(1)} - y_{n+1}^{(0)}\|, \end{aligned}$$

it follows that for sufficiently small h the fixed-point iteration converges, $y_{n+1}^{(\ell)} \rightarrow y_{n+1}$ as $\ell \rightarrow \infty$, provided $h|\beta|L < 1$.

If in each step only a **fixed** number m of correction steps is used, the scheme is called a predictor–corrector method, written $P(EC)^m$: predict with an explicit AB- k step, then evaluate and correct m times. Typical values are $m = 1, 2, 3$ (Moulton, 1926).

End of session 5

II.2 BDF Methods (Backward Differentiation Formulas)

BDF methods differentiate an interpolating polynomial of y -values — rather than integrate one of f -values, as Adams methods do.

Let $q(x)$ be a polynomial of degree $\leq k$ with $q(x_i) = y_i$, $i = n - k + 1, \dots, n, n + 1$ — see Figure II.2. We now need to impose an additional condition to determine y_{n+1} uniquely: y_{n+1} is itself an interpolation node, hence an extra unknown, and none of the conditions $q(x_i) = y_i$ involve f , so $y' = f(x, y)$ has not been used yet.

Idea. *Require that $q(x)$ satisfies the differential equation exactly at $x = x_{n+1}$.*

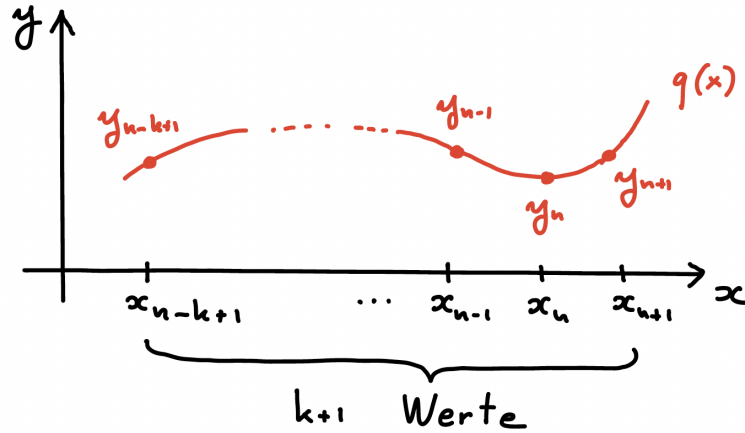


Fig. II.2: Interpolation of the $k + 1$ values $y_{n-k+1}, \dots, y_n, y_{n+1}$ in BDF methods.

From Proposition II.1 with $k + 1$ nodes (similar to AM):

$$q(x) = q(x_n + sh) = \sum_{j=0}^k \binom{s+j-2}{j} \nabla^j y_{n+1}.$$

Now require:

$$\begin{aligned} f(x_{n+1}, y_{n+1}) &= \left. \frac{d}{dx} q(x) \right|_{x=x_{n+1}} = \frac{1}{h} \left. \frac{d}{ds} q(x_n + sh) \right|_{s=1} \\ f(x_{n+1}, y_{n+1}) &= \left. \frac{d}{dx} q(x) \right|_{x=x_{n+1}} = \frac{1}{h} \sum_{j=0}^k \underbrace{\left. \frac{d}{ds} \binom{s+j-2}{j} \right|_{s=1}}_{=0 \text{ for } j=0} \nabla^j y_{n+1} \end{aligned}$$

For $j \geq 1$, the last factor, $(s-1)$, vanishes at $s = 1$ — so by the product rule, only the term that differentiates it survives:

$$\begin{aligned} \left. \frac{d}{ds} \binom{s+j-2}{j} \right|_{s=1} &= \left. \frac{d}{ds} \frac{(s+j-2)(s+j-3) \dots s(s-1)}{j!} \right|_{s=1} \\ &= \left. \frac{(s+j-2)(s+j-3) \dots s}{j!} \right|_{s=1} \\ &= \frac{(j-1)(j-2) \dots 2 \times 1}{j!} = \frac{(j-1)!}{j!} = \frac{1}{j} \end{aligned}$$

BDF- k method:

$$\sum_{j=1}^k \frac{1}{j} \nabla^j y_{n+1} = h f_{n+1}$$

From the above “BDF formula,” we immediately obtain the first BDF- k methods:

- $k = 1$: $y_{n+1} - y_n = hf_{n+1}$ (implicit Euler)
- $k = 2$: $\frac{3}{2}y_{n+1} - 2y_n + \frac{1}{2}y_{n-1} = hf_{n+1}$ (BDF-2)

For $k \geq 1$, a BDF method can, in general, be rewritten (see exercises) as:

$$\kappa_{0,k}y_{n+1} + \kappa_{1,k}y_n + \cdots + \kappa_{k,k}y_{n-k+1} = hf_{n+1}, \quad \text{with} \quad \kappa_{m,k} = (-1)^m \sum_{j=\max\{1,m\}}^k \frac{1}{j} \binom{j}{m}$$

BDF methods are implicit.

II.3 Other Multistep Methods

In the derivation of the Adams methods, the interpolation polynomial $p(t)$ was integrated over $[x_n, x_{n+1}]$. Instead, we can enlarge the integration interval, for example, to $[x_{n-1}, x_{n+1}]$:

$$y(x_{n+1}) = y(x_{n-1}) + \int_{x_{n-1}}^{x_{n+1}} f(t, y(t)) dt \quad (\text{II.3.3})$$

As in the Adams–Bashforth methods, we replace f in (II.3.3) by a polynomial $p(t)$ of degree $\leq k-1$,

$$p(x_i) = f(x_i, y_i), \quad i = n-k+1, \dots, n, \quad k \geq 1,$$

with $x_{i+1} - x_i = h, \forall i$ equally spaced.

As before, we then have

$$\begin{aligned} y_{n+1} &= y_{n-1} + \int_{x_{n-1}}^{x_{n+1}} p(t) dt = y_{n-1} + h \int_{-1}^1 p(x_n + sh) ds \\ &\stackrel{\text{Proposition II.1}}{=} y_{n-1} + h \sum_{j=0}^{k-1} \underbrace{\int_{-1}^1 \binom{s+j-1}{j} ds}_{=:\tilde{\gamma}_j} \nabla^j f_n \end{aligned}$$

This leads to the Nyström Methods.

Nyström Methods (1925).

$$y_{n+1} = y_{n-1} + h \sum_{j=0}^{k-1} \tilde{\gamma}_j \nabla^j f_n \quad (\text{explicit})$$

Remark II.5. For $k = 1$ only the term $j = 0$ survives, and

$$\tilde{\gamma}_0 = \int_{-1}^1 \binom{s-1}{0} ds = \int_{-1}^1 1 ds = 2,$$

which gives $y_{n+1} = y_{n-1} + 2hf_n$ (midpoint method).

Similarly, as in the Adams–Moulton methods, instead of $p(t)$ we can integrate $p^*(t)$ of degree $\leq k$,

$$p^*(x_i) = f(x_i, y_i), \quad i = n-k+1, \dots, n, n+1, \quad k \geq 0$$

over $[x_{n-1}, x_{n+1}]$:

Milne–Simpson Methods (1926).

$$y_{n+1} = y_{n-1} + h \sum_{j=0}^k \tilde{\gamma}_j^* \nabla^j f_{n+1} \quad (\text{implicit except for } k = 1)$$

with

$$\tilde{\gamma}_j^* = \int_{-1}^1 \binom{s+j-2}{j} ds.$$

With $\tilde{\gamma}_0^* = 2$ and $\tilde{\gamma}_1^* = -2$, we recover known methods for $k = 0, 1$. For $k \geq 2$, however, we obtain new methods:

- $k = 0$: $y_{n+1} = y_{n-1} + 2hf_{n+1}$ (implicit Euler with step $2h$, from x_{n-1} to x_{n+1})
- $k = 1$: $y_{n+1} = y_{n-1} + 2hf_n$ (Nyström or midpoint method)
- $k = 2$: $y_{n+1} = y_{n-1} + \frac{h}{3}[f_{n+1} + 4f_n + f_{n-1}]$ (Milne method, Simpson's rule as quadrature)
- etc..

II.4 Order of Multistep Methods

Consider a general linear multistep method (LMM) with k steps

$$\sum_{i=0}^k \alpha_i y_{n+i} = h \sum_{i=0}^k \beta_i f_{n+i}, \quad \alpha_i, \beta_i \in \mathbb{R} \quad (\text{II.4.4})$$

where $f_{n+i} = f(x_{n+i}, y_{n+i})$ and $x_{i+1} - x_i = h > 0, \forall i$ are equidistant.

Moreover, we assume: $\alpha_k \neq 0$ (y_{n+k} is well-defined) and $|\alpha_0| + |\beta_0| \neq 0$ (otherwise it would be a $(k-1)$ -step method).

Remark II.6.

1. In contrast to Sections II.1–II.3, the indices do not run from $n-k+1, \dots, n, n+1$ but from $n, \dots, n+k$, which is always possible by a simple reindexing.
2. If $\beta_k = 0$, then the multistep method is explicit; otherwise, it is implicit.
3. The values $y_0, \dots, y_{k-1}$ (y_0 given) require starting values

$$y_i = \chi_i(h), \quad \lim_{h \rightarrow 0} (\chi_i(h) - y(x_0 + ih)) = 0, \quad i = 0, \dots, k-1,$$

where χ_i is the starting procedure seen as a function of the step size h .

Definition II.2. Let $y(x)$ be a solution of the differential equation $y' = f(x, y)$ and set $y_i := y(x_i)$, $i = n, \dots, n+k-1$. Then the (local) **discretization error** is defined as $y(x_{n+k}) - y_{n+k}$, where y_{n+k} is given by the multistep method.

Remark II.7. This is the numerical error after ONE step under the assumption that the previous k values lie on the exact solution. For $k = 1$, this coincides with the earlier definition for one-step methods.

Definition II.3. A multistep method is called **consistent of (error) order** $p \geq 1$ if the discretization error $y(x_{n+k}) - y_{n+k} = \mathcal{O}(h^{p+1})$, $h \rightarrow 0$, for all differential equations $y' = f(x, y)$ with f sufficiently differentiable.

Theorem II.1. A multistep method with k steps ($i = 0, \dots, k$) is consistent of order $p \geq 1$

$$\iff \sum_{i=0}^k \alpha_i = 0 \quad \text{and} \quad \sum_{i=0}^k \alpha_i i^q = q \sum_{i=0}^k \beta_i i^{q-1}, \quad q = 1, \dots, p$$